

Akash Rawat

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Summary

Senior DevOps & Cloud Engineer with 5+ years of experience designing and operating high-scale, low-latency distributed systems across GCP and AWS. Strong expertise in Kubernetes, infrastructure automation, and observability, with hands-on experience in real-time systems, multi-region architectures, and performance optimization. Solid foundation in Linux systems and networking fundamentals including TCP/IP, load balancing, and system-level tuning for high availability and reliability.

Core Skills

Cloud & Infrastructure: GCP (VPC, IAM, Compute Engine, Cloud Armor, Load Balancing), AWS (EC2, EKS, VPC)

Containers & Orchestration: Kubernetes, Docker, Helm, Service Networking

Networking & Systems: TCP/IP, DNS, HTTP/HTTPS, Load Balancing (L4/L7), NGINX, Linux system tuning (ulimits, process handling), network troubleshooting

Infrastructure as Code: Terraform, Ansible

CI/CD: GitHub Actions, Jenkins, Cloud Build

Observability: Prometheus, Grafana, ELK Stack, OpenTelemetry (basic exposure)

Data & Messaging: Kafka, Redis, RabbitMQ, MySQL, PostgreSQL, MongoDB

Scripting: Python, Bash

Security: IAM, network security, DevSecOps practices

Professional Experience

Emperia Ltd (UK, Remote)

Cloud & DevOps Engineer – GCP & GPU Platform Engineering
Oct 2024 – Dec 2025

- Architected and managed GCP-based infrastructure (VPCs, subnets, IAM, firewall rules) for production-grade, real-time workloads.
 - Designed a custom GPU VM autoscaler and traffic distribution system using load balancing strategies (L4/L7) to support low-latency Pixel Streaming workloads.
 - Built a real-time GPU-based streaming platform with per-user compute isolation, requiring efficient session handling and scalable infrastructure design.
 - Engineered multi-region and multi-zone deployment strategies with traffic-aware routing to improve latency, availability, and failover handling.
 - Optimized system performance by improving resource allocation, reducing latency, and enhancing throughput in distributed workloads.
 - Automated provisioning, health checks, and operational workflows using Python and Bash, reducing manual effort by 20+ hours/week.
 - Implemented observability pipelines (Prometheus, Grafana, logging) to monitor latency, throughput, and system bottlenecks.
 - Built CI/CD pipelines using GitHub Actions, Jenkins, and Cloud Build to improve deployment reliability and speed.
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Omphalos Technology (Remote)

DevOps Engineer – Platform & Delivery

Nov 2023 – Jul 2024

- Led large-scale Git migration for 200+ developers, improving repository governance and operational efficiency.
 - Containerized services using Docker and improved infrastructure efficiency, reducing operational costs by 20%.
 - Designed secure cloud environments using VPCs, IAM, and network-level controls aligned with least-privilege principles.
 - Built CI/CD pipelines for microservices and automated deployment workflows.
 - Deployed ELK-based observability stack to enhance monitoring, logging, and troubleshooting efficiency.
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Omphalos Technology (Remote)

DevOps Engineer (Freelance) – AWS & CI/CD Automation

Oct 2022 – Oct 2023

- Developed end-to-end CI/CD pipelines for AWS-based applications, improving release cadence and rollback capabilities.
- Deployed containerized applications with load balancing and autoscaling to handle dynamic traffic patterns.

- Automated infrastructure provisioning and deployment workflows using scripting and IaC tools.
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Addverb Technologies (Noida, India)

DevOps Engineer – Kubernetes & DevSecOps

Jul 2021 – Sep 2022

- Designed and managed fault-tolerant Kubernetes clusters ensuring high availability (99.99% uptime).
 - Implemented DevSecOps practices, reducing critical vulnerabilities by 40%.
 - Managed Kafka and Zookeeper clusters for distributed messaging systems.
 - Automated build and deployment pipelines, improving efficiency and reliability.
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300 Plus Innovation Solution (Times Internet Client)

DevOps Engineer – EKS & Release Engineering

Feb 2020 – Jul 2021

- Designed scalable CI/CD pipelines, improving release velocity by 30%.
 - Migrated legacy systems to Docker-based architectures, improving portability and consistency.
 - Deployed applications on AWS EKS with Helm, improving rollout reliability.
 - Optimized database performance and implemented schema versioning with Flyway.
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Generic System Solution (Noida, India)

DevOps Associate – Automation & Delivery

Oct 2018 – Feb 2020

- Automated build and deployment workflows using Jenkins and Docker, reducing manual workload by 25%.
 - Developed Python and shell scripts to standardize and streamline release processes.
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Key Projects

Real-Time GPU Streaming Platform (Unreal Pixel Streaming)

- Built scalable, low-latency infrastructure supporting concurrent user sessions with per-user GPU provisioning. - Implemented autoscaling, traffic routing, and cost optimization strategies.

Infrastructure Automation (IaC)

- Developed Infrastructure as Code solutions using Terraform and Ansible to reduce downtime and improve consistency.

Resilient Container Platforms

- Designed Kubernetes-based platforms to improve service availability and reduce downtime by 40%.

Education

MSc Data Science

University of Essex, UK

Oct 2022 – Nov 2023

B.Tech Electronics & Communication Engineering

B K Birla Institute of Engineering & Technology, India

Certifications

- Certified Kubernetes Administrator (CKA)
 - DevOps Certification (Edureka)
 - NGINX Fundamentals
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Additional Highlights

- Experience working with real-time systems and latency-sensitive workloads
- Strong debugging skills in Linux environments
- Hands-on experience with distributed systems and microservices architectures